

Calculus I Lesson 34B

1) $f(x) = x + 3$ $0 \leq x \leq 5$

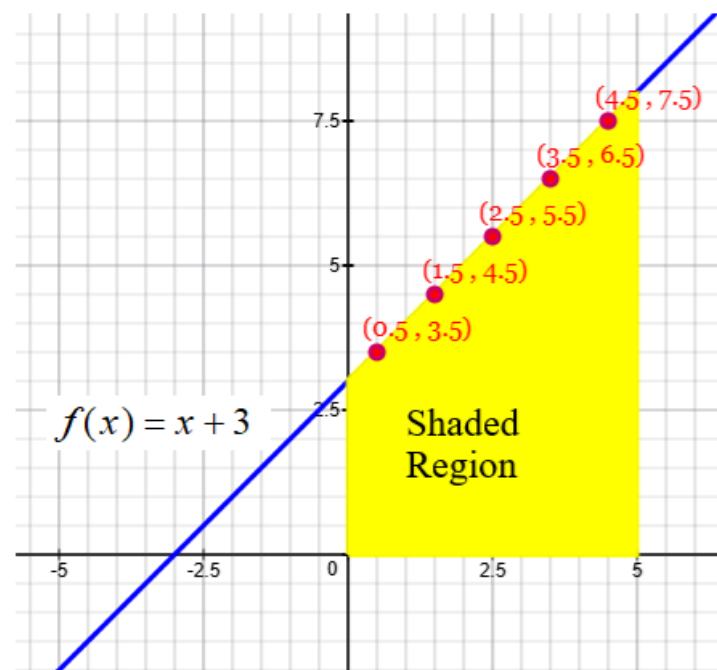
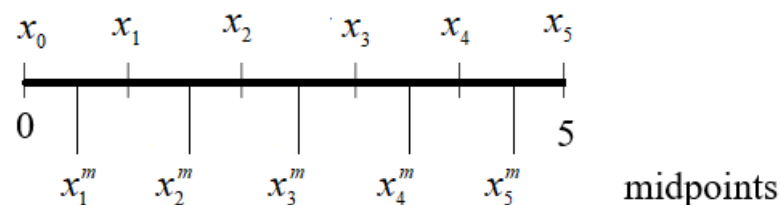
Find the shaded area between the x -axis and $f(x)$.

Let $a = 0$ and $b = 5$

Let $n =$ number of subintervals $= 5$.

Let $\Delta x = \frac{b-a}{n} = \frac{5-0}{5} = 1$ (Note: Δx is the width of each rectangle.)

Partition the interval $[a, b]$ as follows: $a = x_0 < x_1 < x_2 < x_3 < x_4 < x_5 = b$



a) Height of each rectangle:

subinterval 1 is from $x_0 = 0$ to $x_1 = 1$; midpoint $= x_1^m = 0.5$; $f(x_1^m) = \underline{\hspace{1cm}}$

subinterval 2 is from $x_1 = 1$ to $x_2 = 2$; midpoint $= x_2^m = 1.5$; $f(x_2^m) = \underline{\hspace{1cm}}$

subinterval 3 is from $x_2 = 1$ to $x_3 = 3$; midpoint $= x_3^m = 2.5$; $f(x_3^m) = \underline{\hspace{1cm}}$

subinterval 4 is from $x_3 = 1$ to $x_4 = 4$; midpoint $= x_4^m = 3.5$; $f(x_4^m) = \underline{\hspace{1cm}}$

subinterval 5 is from $x_4 = 1$ to $x_5 = 5$; midpoint $= x_5^m = 4.5$; $f(x_5^m) = \underline{\hspace{1cm}}$

b) Height of each rectangle:

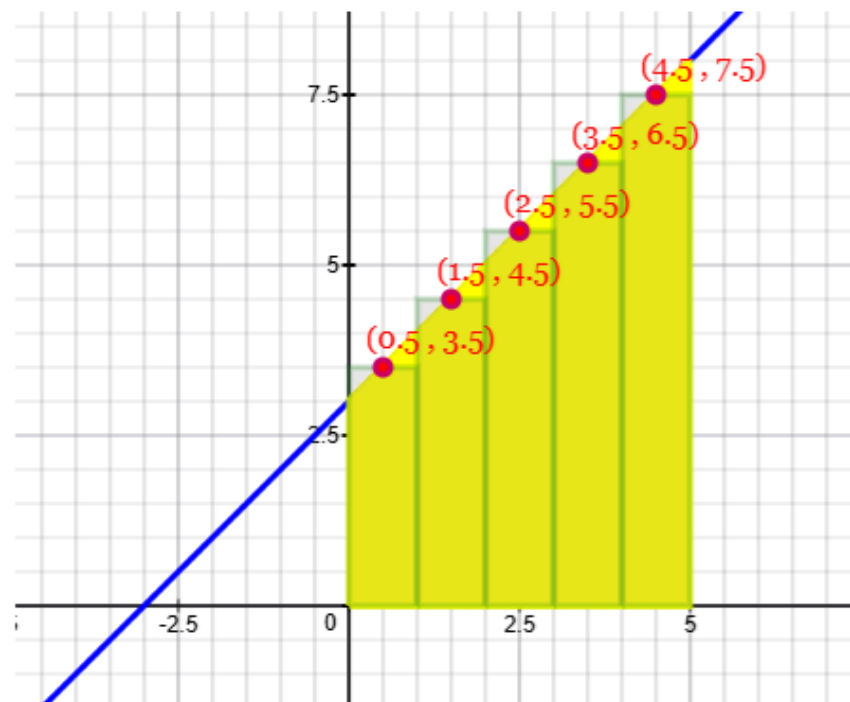
$A_1 = \text{Area of rectangle 1} = f(x_1^m) \cdot \Delta x = \underline{\hspace{1cm}}$

$A_2 = \text{Area of rectangle 2} = f(x_2^m) \cdot \Delta x = \underline{\hspace{1cm}}$

$A_3 = \text{Area of rectangle 3} = f(x_3^m) \cdot \Delta x = \underline{\hspace{1cm}}$

$A_4 = \text{Area of rectangle 4} = f(x_4^m) \cdot \Delta x = \underline{\hspace{1cm}}$

$A_5 = \text{Area of rectangle 5} = f(x_5^m) \cdot \Delta x = \underline{\hspace{1cm}}$



c) Sum of areas of rectangles

$$= A_1 + A_2 + A_3 + A_4 + A_5$$

$$= f(x_1^m) \cdot \Delta x + f(x_2^m) \cdot \Delta x + f(x_3^m) \cdot \Delta x + f(x_4^m) \cdot \Delta x + f(x_5^m) \cdot \Delta x$$

$$= \sum_{i=1}^5 f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}}$$

d) Area of shaded region is approximately ? .

2) $f(x) = -2x + 4$ $0 \leq x \leq 2$

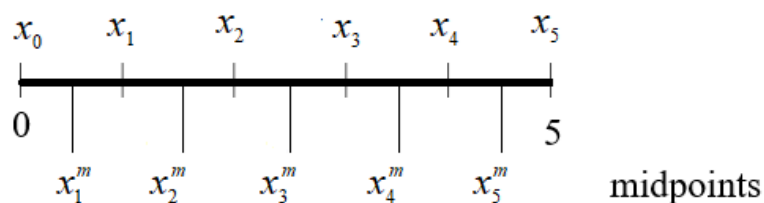
Find the shaded area between the x -axis and $f(x)$.

Let $a = 0$ and $b = 2$

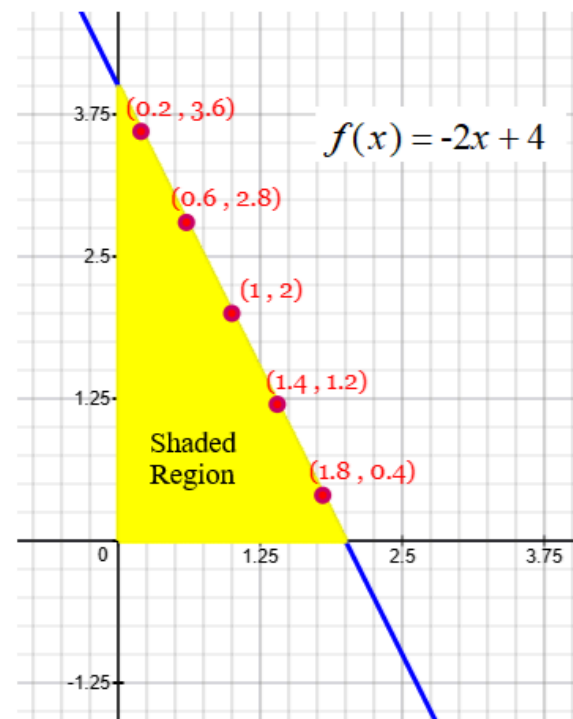
Let $n =$ number of subintervals $= 5$.

Let $\Delta x = \frac{b-a}{n} = \frac{2-0}{5} = 0.4$ (Note: Δx is the width of each rectangle.)

Partition the interval $[a, b]$ as follows: $a = x_0 < x_1 < x_2 < x_3 < x_4 < x_5 = b$



Note: $x_0 = 0$; $x_1 = 1 \cdot \Delta x$; $x_2 = 2 \cdot \Delta x$; $x_3 = 3 \cdot \Delta x$; ... $x_i = i \cdot \Delta x$



a) Height of each rectangle:

subinterval 1 is from $x_0 = 0$ to $x_1 = 0.4$; midpoint = $x_1^m = 0.2$; $f(x_1^m) = \underline{\hspace{1cm}}$

subinterval 2 is from $x_1 = 0.4$ to $x_2 = 0.8$; midpoint = $x_2^m = 0.6$; $f(x_2^m) = \underline{\hspace{1cm}}$

subinterval 3 is from $x_2 = 0.8$ to $x_3 = 1.2$; midpoint = $x_3^m = 1$; $f(x_3^m) = \underline{\hspace{1cm}}$

subinterval 4 is from $x_3 = 1.2$ to $x_4 = 1.6$; midpoint = $x_4^m = 1.4$; $f(x_4^m) = \underline{\hspace{1cm}}$

subinterval 5 is from $x_4 = 1.6$ to $x_5 = 2$; midpoint = $x_5^m = 1.8$; $f(x_5^m) = \underline{\hspace{1cm}}$

b) Height of each rectangle:

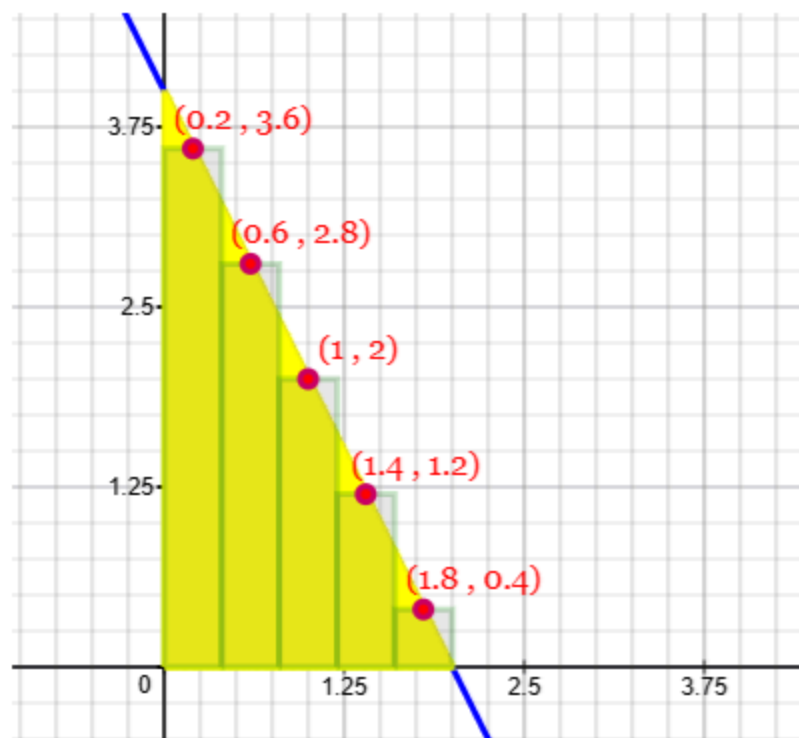
$$A_1 = \text{Area of rectangle 1} = f(x_1^m) \cdot \Delta x = \underline{\hspace{1cm}}$$

$$A_2 = \text{Area of rectangle 2} = f(x_2^m) \cdot \Delta x = \underline{\hspace{1cm}}$$

$$A_3 = \text{Area of rectangle 3} = f(x_3^m) \cdot \Delta x = \underline{\hspace{1cm}}$$

$$A_4 = \text{Area of rectangle 4} = f(x_4^m) \cdot \Delta x = \underline{\hspace{1cm}}$$

$$A_5 = \text{Area of rectangle 5} = f(x_5^m) \cdot \Delta x = \underline{\hspace{1cm}}$$



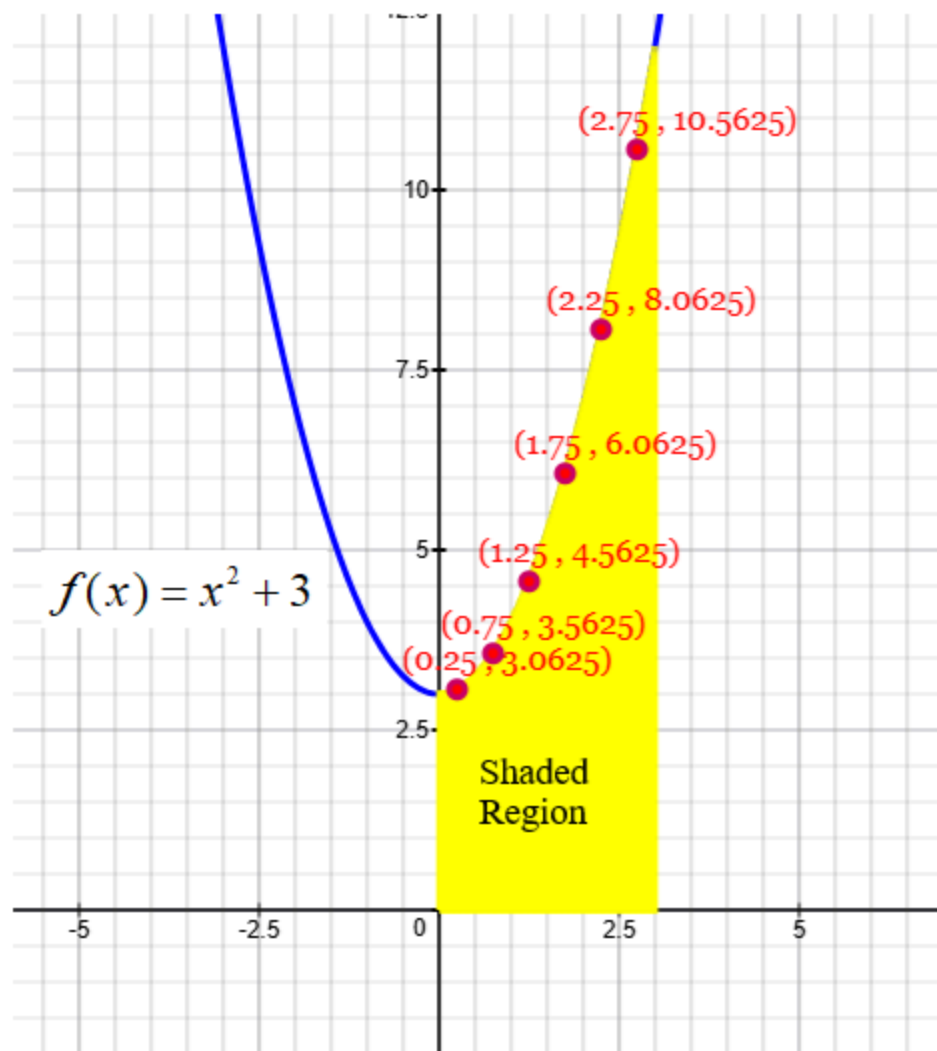
c) Sum of areas of rectangles

$$= A_1 + A_2 + A_3 + A_4 + A_5$$

$$= f(x_1^m) \cdot \Delta x + f(x_2^m) \cdot \Delta x + f(x_3^m) \cdot \Delta x + f(x_4^m) \cdot \Delta x + f(x_5^m) \cdot \Delta x$$

$$= \sum_{i=1}^5 f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}} ?$$

d) Area of shaded region is approximately ? .



a) Height of each rectangle:

subinterval 1 is from $x_0 = 0$ to $x_1 = 0.5$; midpoint $= x_1^m = 0.25$; $f(x_1^m) = \underline{\hspace{1cm}}$

subinterval 2 is from $x_1 = 0.5$ to $x_2 = 1$; midpoint $= x_2^m = 0.75$; $f(x_2^m) = \underline{\hspace{1cm}}$

subinterval 3 is from $x_2 = 1$ to $x_3 = 1.5$; midpoint $= x_3^m = 1.25$; $f(x_3^m) = \underline{\hspace{1cm}}$

subinterval 4 is from $x_3 = 1.5$ to $x_4 = 2$; midpoint $= x_4^m = 1.75$; $f(x_4^m) = \underline{\hspace{1cm}}$

subinterval 5 is from $x_4 = 2$ to $x_5 = 2.5$; midpoint $= x_5^m = 2.25$; $f(x_5^m) = \underline{\hspace{1cm}}$

subinterval 6 is from $x_5 = 2.5$ to $x_6 = 3$; midpoint $= x_5^m = 2.75$; $f(x_5^m) = \underline{\hspace{1cm}}$

b) Height of each rectangle:

$$A_1 = \text{Area of rectangle 1} = f(x_1^m) \cdot \Delta x = \underline{\hspace{1cm}}$$

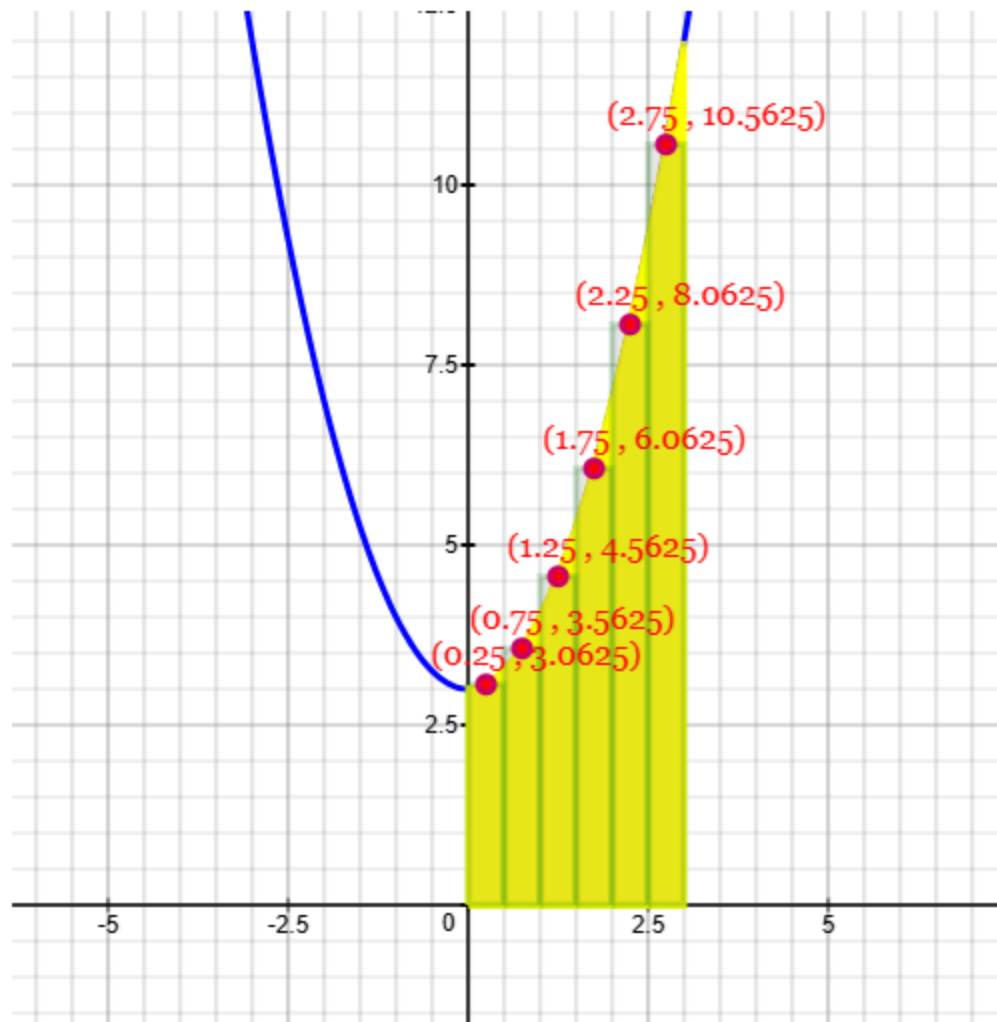
$$A_2 = \text{Area of rectangle 2} = f(x_2^m) \cdot \Delta x = \underline{\hspace{1cm}}$$

$$A_3 = \text{Area of rectangle 3} = f(x_3^m) \cdot \Delta x = \underline{\hspace{1cm}}$$

$$A_4 = \text{Area of rectangle 4} = f(x_4^m) \cdot \Delta x = \underline{\hspace{1cm}}$$

$$A_5 = \text{Area of rectangle 5} = f(x_5^m) \cdot \Delta x = \underline{\hspace{1cm}}$$

$$A_6 = \text{Area of rectangle 6} = f(x_6^m) \cdot \Delta x = \underline{\hspace{1cm}}$$



d) Area of shaded region is approximately ____?

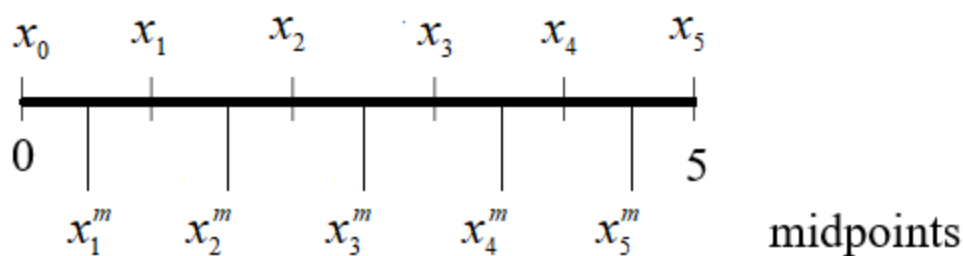
4) $f(x) = 0.5x^3 + 1$ $0 \leq x \leq 2$

Find the shaded area between the x -axis and $f(x)$.

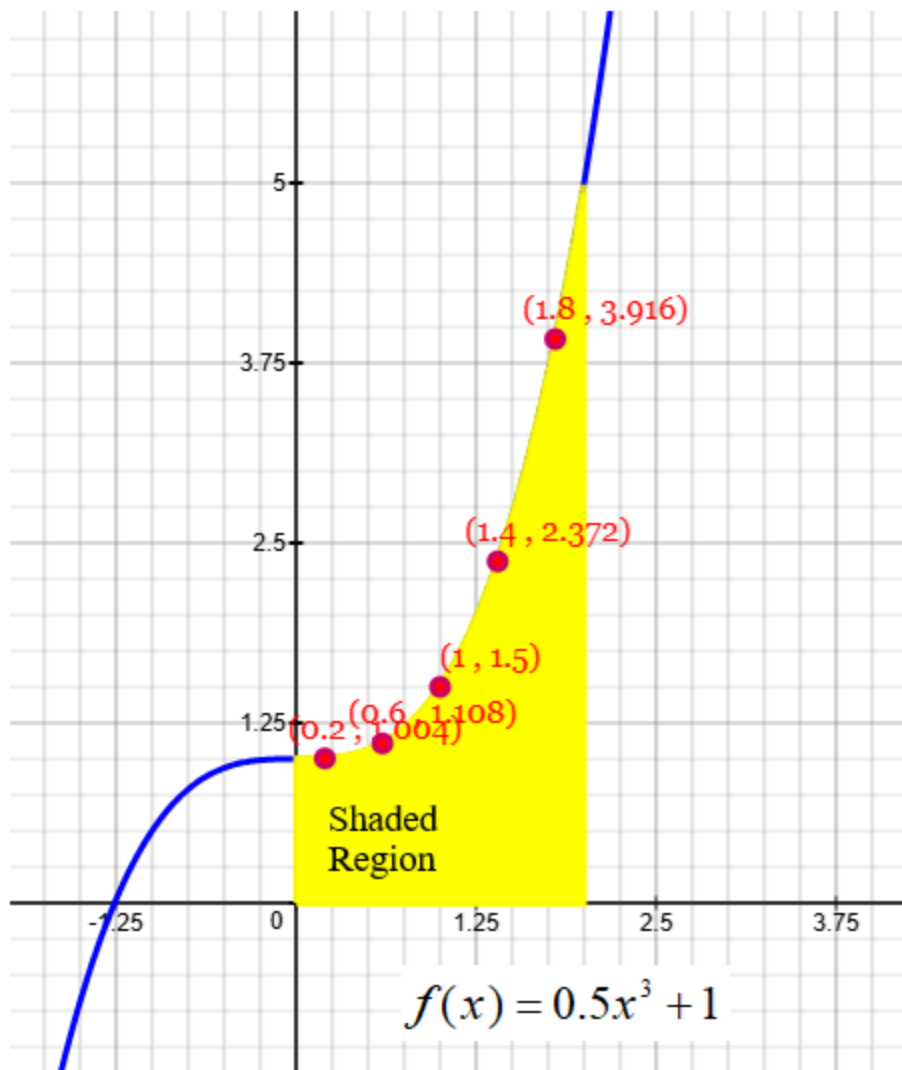
Let $a = 0$ and $b = 2$ Let $n =$ number of subintervals $= 5$.

Let $\Delta x = \frac{b-a}{n} = \frac{2-0}{5} = 0.4$ (Note: Δx is the width of each rectangle.)

Partition the interval $[a, b]$ as follows: $a = x_0 < x_1 < x_2 < x_3 < x_4 < x_5 = b$



Note: $x_0 = 0$; $x_1 = 1 \cdot \Delta x$; $x_2 = 2 \cdot \Delta x$; $x_3 = 3 \cdot \Delta x$; ... $x_i = i \cdot \Delta x$



a) Height of each rectangle:

subinterval 1 is from $x_0 = 0$ to $x_1 = 0.4$; midpoint $= x_1^m = 0.2$; $f(x_1^m) = \underline{\hspace{1cm}}$

subinterval 2 is from $x_1 = 0.4$ to $x_2 = 0.8$; midpoint $= x_2^m = 0.6$; $f(x_2^m) = \underline{\hspace{1cm}}$

subinterval 3 is from $x_2 = 0.8$ to $x_3 = 1.2$; midpoint $= x_3^m = 1$; $f(x_3^m) = \underline{\hspace{1cm}}$

subinterval 4 is from $x_3 = 1.2$ to $x_4 = 1.6$; midpoint $= x_4^m = 1.4$; $f(x_4^m) = \underline{\hspace{1cm}}$

subinterval 5 is from $x_4 = 1.6$ to $x_5 = 2$; midpoint $= x_5^m = 1.8$; $f(x_5^m) = \underline{\hspace{1cm}}$

b) Height of each rectangle:

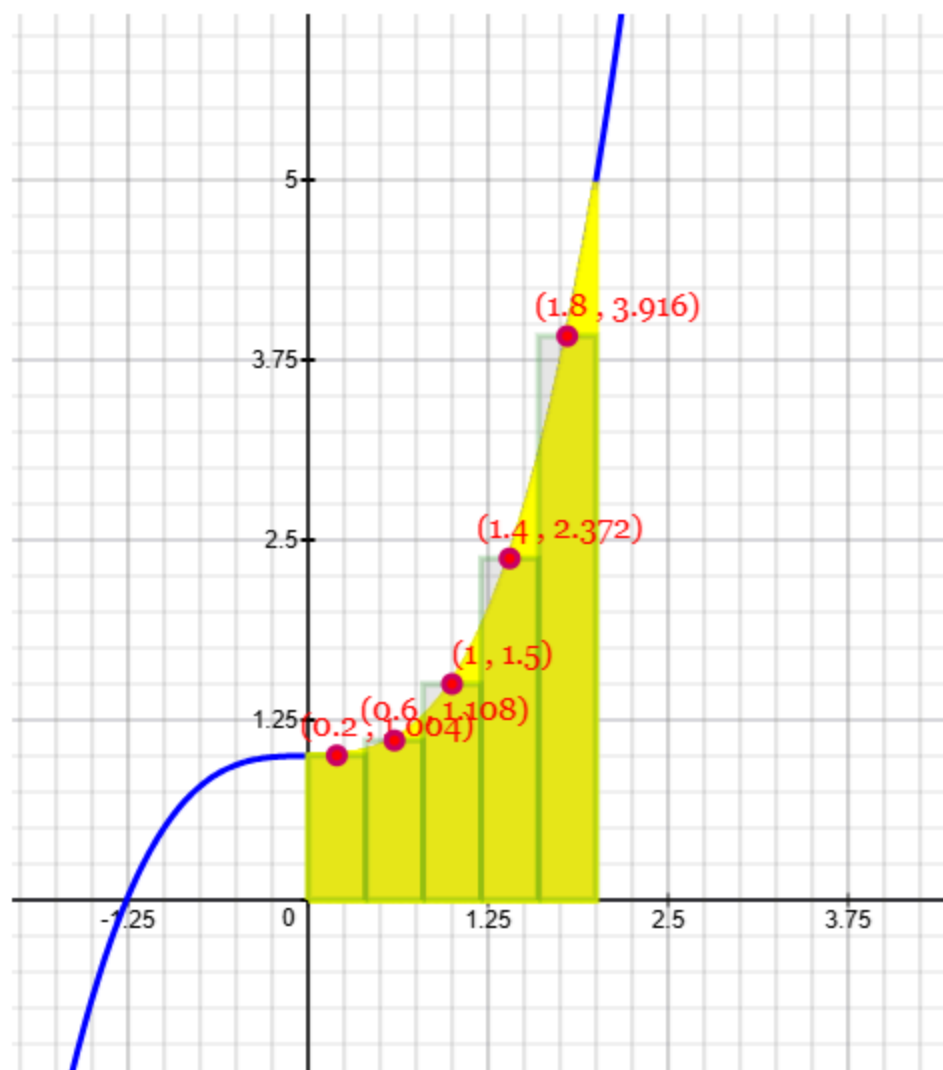
$$A_1 = \text{Area of rectangle 1} = f(x_1^m) \cdot \Delta x = \underline{\hspace{1cm}}$$

$$A_2 = \text{Area of rectangle 2} = f(x_2^m) \cdot \Delta x = \underline{\hspace{1cm}}$$

$$A_3 = \text{Area of rectangle 3} = f(x_3^m) \cdot \Delta x = \underline{\hspace{1cm}}$$

$$A_4 = \text{Area of rectangle 4} = f(x_4^m) \cdot \Delta x = \underline{\hspace{1cm}}$$

$$A_5 = \text{Area of rectangle 5} = f(x_5^m) \cdot \Delta x = \underline{\hspace{1cm}}$$



c) Sum of areas of rectangles

$$= A_1 + A_2 + A_3 + A_4 + A_5$$

$$= f(x_1^m) \cdot \Delta x + f(x_2^m) \cdot \Delta x + f(x_3^m) \cdot \Delta x + f(x_4^m) \cdot \Delta x + f(x_5^m) \cdot \Delta x$$

$$= \sum_{i=1}^5 f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}}$$

d) Area of shaded region is approximately ? .

$$5) f(x) = \sqrt{x} \quad 0 \leq x \leq 5$$

Find the shaded area between the x -axis and $f(x)$.

Let $a = 0$ and $b = 5$

a) Let $n =$ number of subintervals $= 5$.

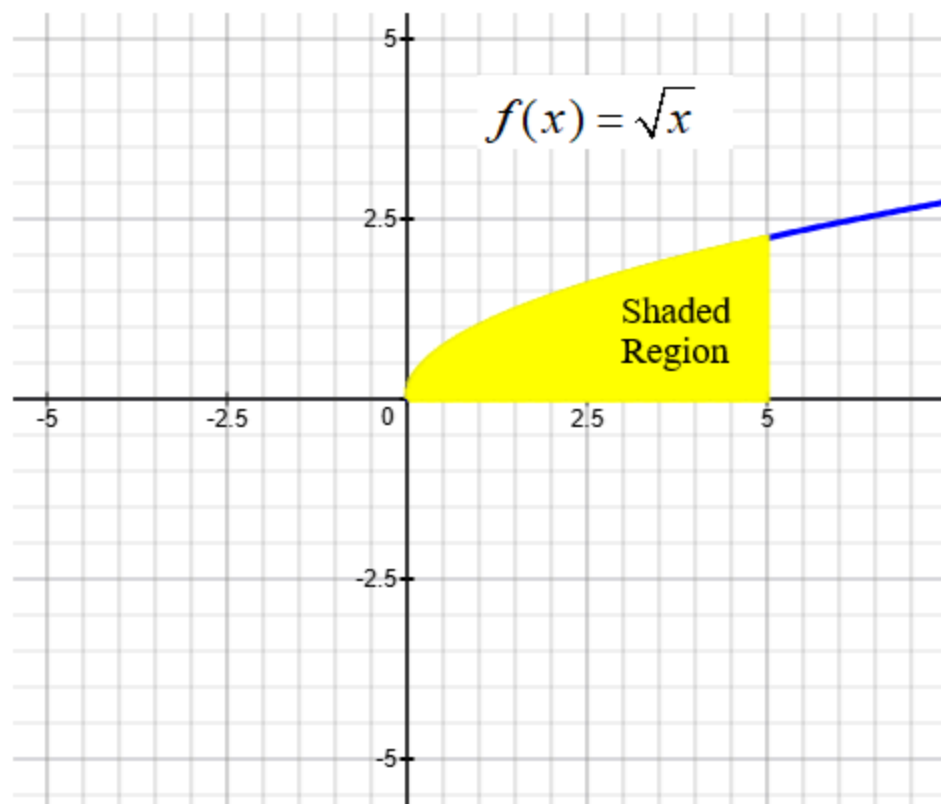
$$\Delta x = \frac{b-a}{n} = \frac{5-0}{5} = 1 \quad (\text{Note: } \Delta x \text{ is the width of each rectangle.})$$

$$\text{Sum of areas of rectangles} = A_1 + A_2 + A_3 + A_4 + A_5$$

$$= f(x_1^m) \cdot \Delta x + f(x_2^m) \cdot \Delta x + f(x_3^m) \cdot \Delta x + f(x_4^m) \cdot \Delta x + f(x_5^m) \cdot \Delta x$$

$$= \sum_{i=1}^5 f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}} ?$$

$$\text{Area of shaded region} \approx \sum_{i=1}^5 f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}} ?$$



b) Let $n =$ number of subintervals $= 50$.

$$\Delta x = \frac{b-a}{n} = \frac{5-0}{50} = 0.1 \quad (\text{Note: } \Delta x \text{ is the width of each rectangle.})$$

$$\text{Area of shaded region} \approx \sum_{i=1}^{50} f(x_i^m) \cdot \Delta x = \underline{\hspace{1cm}}.$$

c) Let $n =$ number of subintervals $= 5,000$.

$$\Delta x = \frac{b-a}{n} = \frac{5-0}{5000} = 0.001 \quad (\text{Note: } \Delta x \text{ is the width of each rectangle.})$$

$$\text{Area of shaded region} \approx \sum_{i=1}^{5000} f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}}.$$

$$6) f(x) = x + \sin x \quad 0 \leq x \leq 2\pi$$

Find the shaded area between the x -axis and $f(x)$.

$$\text{Let } a = 0 \quad \text{and} \quad b = 2\pi$$

a) Let n = number of subintervals = 10.

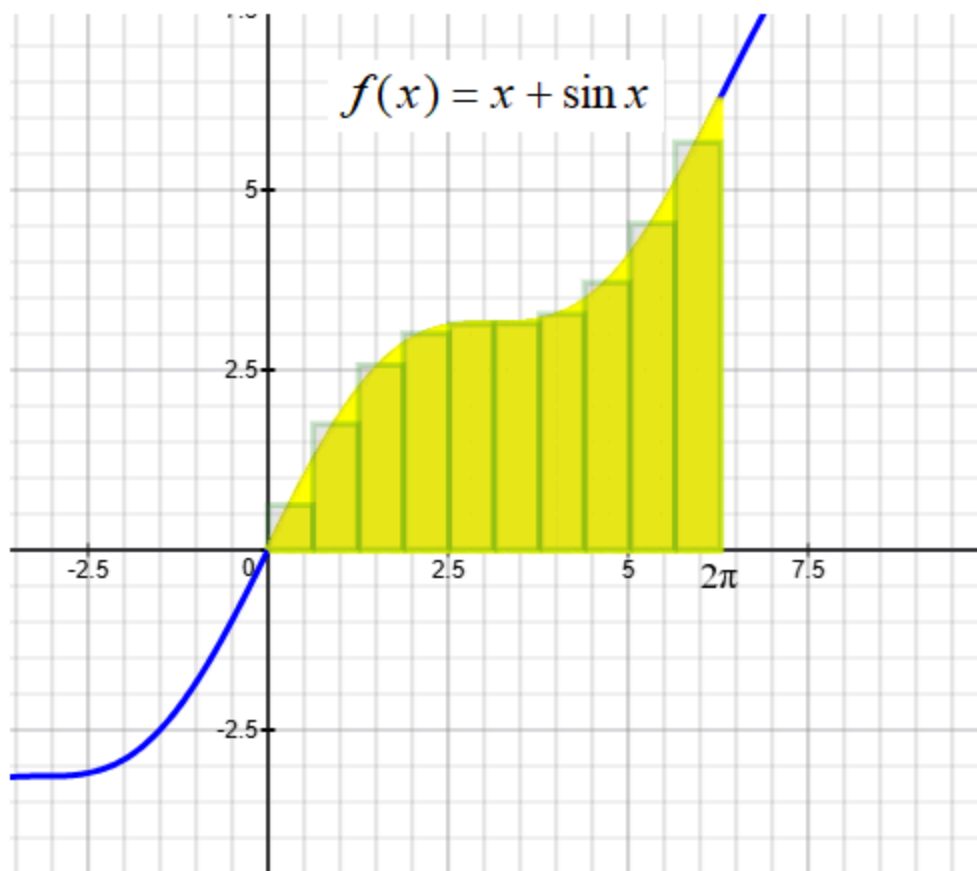
$$\Delta x = \frac{b-a}{n} = \frac{2\pi - 0}{10} = \frac{\pi}{5} \quad (\text{Note: } \Delta x \text{ is the width of each rectangle.})$$

$$\text{Sum of areas of rectangles} = A_1 + A_2 + A_3 + A_4 + A_5 + \dots + A_{10}$$

$$= f(x_1^m) \cdot \Delta x + f(x_2^m) \cdot \Delta x + f(x_3^m) \cdot \Delta x + f(x_4^m) \cdot \Delta x + \dots + f(x_{10}^m) \cdot \Delta x$$

$$= \sum_{i=1}^{10} f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}} ?$$

$$\text{Area of shaded region} \approx \sum_{i=1}^{10} f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}} ?$$



b) Let $n =$ number of subintervals $= 100$.

$$\Delta x = \frac{b-a}{n} = \frac{2\pi - 0}{100} \quad (\text{Note: } \Delta x \text{ is the width of each rectangle.})$$

$$\text{Area of shaded region} \approx \sum_{i=1}^{100} f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}}.$$

c) Let $n =$ number of subintervals $= 1,000$.

$$\Delta x = \frac{b-a}{n} = \frac{2\pi - 0}{1000} \quad (\text{Note: } \Delta x \text{ is the width of each rectangle.})$$

$$\text{Area of shaded region} \approx \sum_{i=1}^{1000} f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}}.$$

$$7) f(x) = -x - \cos x \quad 0 \leq x \leq 2\pi$$

Find the shaded area between the x -axis and $f(x)$.

Let $a = 0$ and $b = 2\pi$

a) Let $n =$ number of subintervals $= 10$.

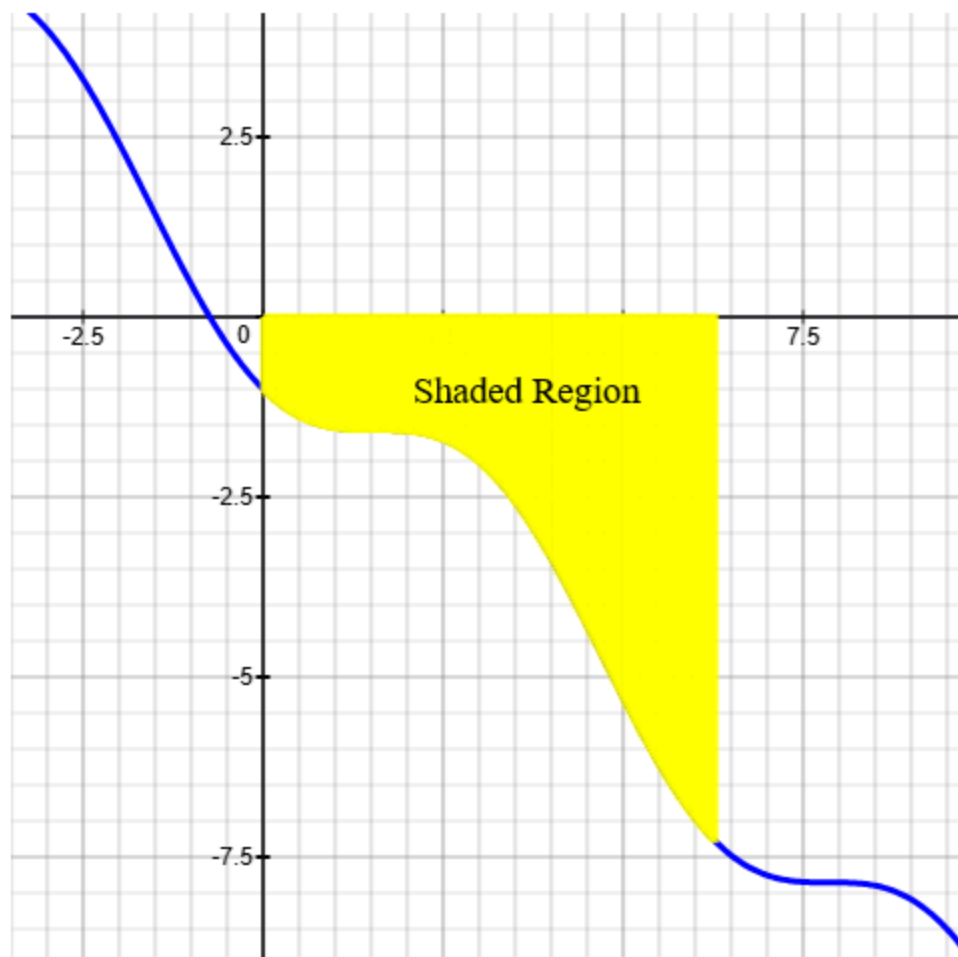
$$\Delta x = \frac{b-a}{n} = \frac{2\pi - 0}{10} = \frac{\pi}{5} \quad (\text{Note: } \Delta x \text{ is the width of each rectangle.})$$

$$\text{Sum of areas of rectangles} = A_1 + A_2 + A_3 + A_4 + A_5 + \dots + A_{10}$$

$$= f(x_1^m) \cdot \Delta x + f(x_2^m) \cdot \Delta x + f(x_3^m) \cdot \Delta x + f(x_4^m) \cdot \Delta x + \dots + f(x_{10}^m) \cdot \Delta x$$

$$= \sum_{i=1}^{10} f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}}?$$

$$\text{Area of shaded region} \approx \sum_{i=1}^{10} f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}}?.$$



b) Let $n =$ number of subintervals $= 100$.

$$\Delta x = \frac{b-a}{n} = \frac{2\pi-0}{100} \quad (\text{Note: } \Delta x \text{ is the width of each rectangle.})$$

$$\text{Area of shaded region} \approx \sum_{i=1}^{100} f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}}.$$

c) Let $n =$ number of subintervals $= 1,000$.

$$\Delta x = \frac{b-a}{n} = \frac{2\pi-0}{1000} \quad (\text{Note: } \Delta x \text{ is the width of each rectangle.})$$

$$\text{Area of shaded region} \approx \sum_{i=1}^{1000} f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}}.$$

$$8) f(x) = -0.2x^3 - 2 \quad 0 \leq x \leq 3$$

Find the shaded area between the x -axis and $f(x)$.

Let $a = 0$ and $b = 3$

a) Let $n =$ number of subintervals $= 10$.

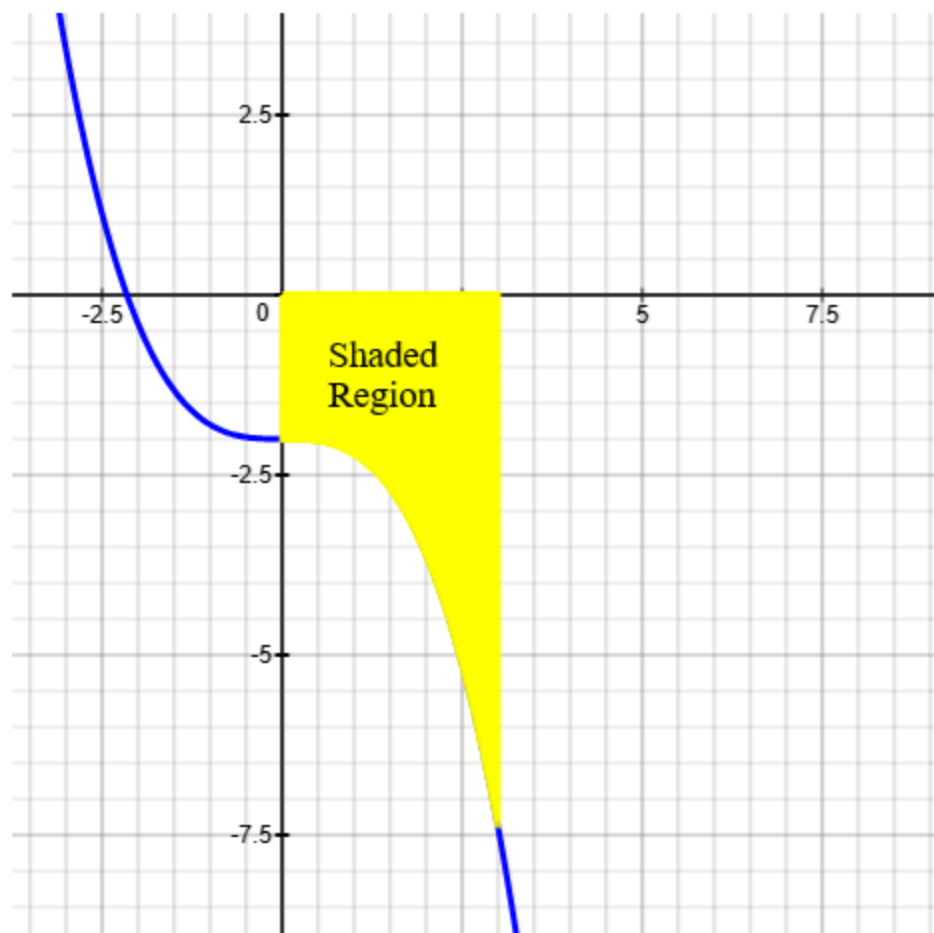
$$\Delta x = \frac{b-a}{n} = \frac{3-0}{10} = 0.3 \quad (\text{Note: } \Delta x \text{ is the width of each rectangle.})$$

$$\text{Sum of areas of rectangles} = A_1 + A_2 + A_3 + A_4 + A_5 + \dots + A_{10}$$

$$= f(x_1^m) \cdot \Delta x + f(x_2^m) \cdot \Delta x + f(x_3^m) \cdot \Delta x + f(x_4^m) \cdot \Delta x + \dots + f(x_{10}^m) \cdot \Delta x$$

$$= \sum_{i=1}^{10} f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}} ?$$

$$\text{Area of shaded region} \approx \sum_{i=1}^{10} f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}} ?$$



b) Let $n =$ number of subintervals $= 100$.

$$\Delta x = \frac{b-a}{n} = \frac{3-0}{100} = 0.03 \quad (\text{Note: } \Delta x \text{ is the width of each rectangle.})$$

$$\text{Area of shaded region} \approx \sum_{i=1}^{100} f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}}.$$

c) Let $n =$ number of subintervals $= 1,000$.

$$\Delta x = \frac{b-a}{n} = \frac{3-0}{1000} = 0.003 \quad (\text{Note: } \Delta x \text{ is the width of each rectangle.})$$

$$\text{Area of shaded region} \approx \sum_{i=1}^{1000} f(x_i^m) \cdot \Delta x = \underline{\hspace{2cm}}.$$

